

Normal Parent-Hamiltonian Range Reduction in Cirac–Perez-Garcia–Schuch–Verstraete 2021

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1 The assertion

The range-reduction step for normal matrix product states appears in Section IV.C of Cirac–Pérez-García–Schuch–Verstraete [CPGSV21].¹

Let A be a normal MPS tensor, and let $L_0 > 0$ be its injectivity length: after blocking L_0 consecutive sites, the resulting tensor is injective. The source first proves an intersection step for two overlapping local Hamiltonian terms of range $L_0 + 1$. These terms act on sites $1, \dots, L_0 + 1$ and $2, \dots, L_0 + 2$. The source denotes the three regions by A , B , and C : the first site, the middle region consisting of sites $2, \dots, L_0 + 1$, and the last site. The tensor associated to the middle region B is injective. For a state in the intersection

$$\mathcal{G}_{1,\dots,L_0+1} \cap \mathcal{G}_{2,\dots,L_0+2},$$

the argument inverts the left boundary tensor A_{left} , restores the middle region B , and then uses injectivity of the enlarged region to invert the joint region. It concludes that every vector in the intersection already lies in $\mathcal{G}_{1,\dots,L_0+2}$; the reverse inclusion is immediate.²

The source then says that this argument can be iterated. Hence the ground space obtained from all length- $L_0 + 1$ terms agrees with the ground space obtained from all longer length- $L_0 + k$ terms. When $k = L_0$, the proof either appeals to the previous $2L_0$ -range uniqueness theorem or applies the same kind of argument when closing the periodic boundary. The resulting theorem states that a normal MPS which becomes injective after blocking L_0 sites has a unique ground state for the parent Hamiltonian whose interaction range is $L_0 + 1$.³

¹For traceability, this note corresponds to issue #1042. The principal formal obligation is recorded in issue #588; the boundary-closing comparison is recorded in issue #2368; and the parent-Hamiltonian consequences are recorded in issue #460.

²Source location: `Papers/2011.12127/TN-Review-main.tex:2049–2077`, especially the middle-region argument around the displayed figures following `eq:4:initiate-intersection-proof`.

³Source location: `Papers/2011.12127/TN-Review-main.tex:2078–2079` for the iteration and boundary-closing sentence, and `Papers/2011.12127/TN-Review-main.tex:2087–2090` for the theorem labeled `thm:4:unique-gs-L0_plus_1`.

2 The formal statement

The corresponding formal statement keeps the intended hypotheses. In mathematical terms, it assumes that A is normal, that A is injective after blocking a positive number L_0 of sites, and that the periodic chain has length $N \geq 2$ with an interaction range L satisfying $L_0 < L \leq N$. The desired conclusion is that the periodic-chain ground space is the line spanned by the periodic MPS vector $\Omega_N(A)$:

$$\mathcal{G}_{N,L}(A) = \mathbb{C} \Omega_N(A).$$

In the formal statement this is the equality between the cyclic chain ground space and the subspace $\mathbb{C} \Omega_N(A)$. The hard inclusion into $\mathbb{C} \Omega_N(A)$ is the remaining range-reduction lemma.⁴

The blueprint records the same range-reduction target in the theorem “Normal-range reduction: containment in the MPS span” in `blueprint/src/chapter/ch14_parent_hamiltonian.tex:1648–1686`, labelled `thm:normal_range_reduction_containment`. Its proof sketch already separates the open-chain containment from the boundary-closing comparison, which is the same division made by the formal statements below.

The first mismatch is that the available intersection theorem is not a literal expression of the proof through the region B . That theorem was written for a tensor that is already injective at one site. Its proof decomposes the identity as a linear combination of the one-site matrices A_j and then iterates this single-letter intersection property through contiguous windows.⁵ For a normal tensor with injectivity length L_0 , the available inverse is the inverse for words of length L_0 , not a one-site inverse. This is not a contradiction in the source argument; it is a mismatch between the diagrammatic region-by-region proof and the available formal statements for intersection properties.

The formal argument therefore separates the proof into two parts instead of reproducing the compressed source proof verbatim. The open-chain component has been proved by a block-injective range-reduction lemma: if every contiguous window of length $L_0 + 1$ lies in the corresponding local MPS ground space, then the full open chain lies in the full MPS ground space. Combining this with cyclic window monotonicity gives an open-chain consequence from the cyclic length- L constraints. This supplies the formal analog of the iterated open-boundary part of the source proof, without expressing the inverse for the region B through the older single-site intersection lemma.⁶

⁴Formal locations: `TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean:836` for `MPSTensor.chainGroundSpace_le_mpvSubmodule_of_normal_range_reduction`, and `TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean:872` for `MPSTensor.chainGroundSpace_eq_mpvSubmodule_normal`. The formal hypotheses are named `IsNormalA` and `IsNBlkInjectiveAL0`, together with $0 < L_0$, $2 \leq N$, and $L_0 < L \leq N$.

⁵Formal locations: `TNLean/MPS/ParentHamiltonian/IntersectionProperty.lean:281–284` for `MPSTensor.groundSpace_intersection`, `TNLean/MPS/ParentHamiltonian/IntersectionProperty.lean:341–364` for the use of `decompositionMaphA1`, and `TNLean/MPS/ParentHamiltonian/CyclicWindow.lean:153–163` for the contiguous iteration using that theorem.

⁶Formal locations: `TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean:266` for `MPSTensor.contiguous_mem_groundSpace_of_isNBlkInjective`, and `TNLean/MPS/ParentHam`

3 The remaining comparison

After the open-chain step, a periodic-chain ground state can be written with an open-chain boundary matrix X . The point left to prove is that the cyclic length- $L_0 + 1$ constraints force X to be scalar, and hence that the state is in $\mathbb{C} \Omega_N(A)$. The formal proof has extracted the two one-sided cyclic-window compatibilities at the boundary. In schematic notation, the last-site cyclic window used in closing the boundary gives

$$C_\tau^+ A_j X = Y_\tau^+ A_j,$$

while the opposite cyclic window used in closing the boundary gives

$$X A_j C_\tau^- = A_j Y_\tau^-.$$

Here τ denotes the complementary boundary word exposed by such a cyclic window, equivalently the physical labels outside the open middle interval. The formal cyclic-window lemmas produce analogous complement-word products, written schematically as C_τ^+ and C_τ^- , but the two products are obtained with different orientations and indexings. The two formulas are both valid; what is missing is a comparison that identifies the two boundary matrices after the complementary sites have been indexed by the same word.⁷

The last algebraic step needs the two one-sided identities to use the same matrix. There should be a word μ on the complementary sites and matrices Y_μ such that, for every physical letter j ,

$$A^\mu A_j X = Y_\mu A_j, \quad X A_j A^\mu = A_j Y_\mu.$$

Once these two identities use the same complementary-site word μ and the same matrix Y_μ , the formal algebraic lemma proves that X commutes with all words of a fixed positive length; block injectivity then promotes this to commutation with the full matrix algebra, so X is scalar.⁸ The remaining point is the comparison between the two boundary-crossing outputs: one must reindex their complementary words in the same way and identify the two boundary matrices as the same matrix Y_μ . For an N -site vector ψ , write

$$\text{Res}_{i,L}^\tau(\psi)(\sigma) = \psi(\text{replaceWindow}_{L,i}(\sigma, \tau)),$$

where the cyclic window of length L beginning at i is fixed to the word τ , and σ labels the remaining $N - L$ sites. With $N = M + 1$, the two boundary-crossing

iltonian/UniqueGroundState.lean:328 for MPSTensor.chainGroundSpace_le_groundSpace_of_isNBlkInjective.

⁷Formal locations: TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean:441 for MPSTensor.chainGroundSpace_wrapped_boundary_compatibility_of_isNBlkInjective, TNLean/MPS/ParentHamiltonian/WrappingWindow.lean:402 for the wrapped compatibility, and TNLean/MPS/ParentHamiltonian/WrappingWindow.lean:450 for the mirror compatibility.

⁸Formal locations: TNLean/MPS/ParentHamiltonian/WrappingWindow.lean:645 for MPSTensor.commutes_words_of_two_sided_middle_compatibility, and TNLean/MPS/ParentHamiltonian/UniqueGroundState.lean:554 for MPSTensor.groundSpaceMap_mem_mpvSubmodule_of_isNBlkInjective_of_two_sided_middle_compatibility.

words are

$$\tau_{\eta}^{+}(\mu)_i = \begin{cases} \mu_{i-L_0}, & L_0 \leq i < N-1, \\ \eta, & \text{otherwise,} \end{cases} \quad \tau_{\eta}^{-}(\mu)_i = \begin{cases} \mu_{i-1}, & 1 \leq i < N-L_0, \\ \eta, & \text{otherwise.} \end{cases}$$

Equivalently, when closing the boundaries, the closure property should identify the two restrictions

$$\text{Res}_{M,L_0+1}^{\tau_{\eta}^{+}(\mu)}(\psi) = \text{Res}_{M+1-L_0,L_0+1}^{\tau_{\eta}^{-}(\mu)}(\psi),$$

before taking the first-site restriction $\sigma_1 = j$.⁹

4 Conclusion

The classification is an open gap. It identifies a difference between the compressed source proof and the available formal lemmas; it is not a source-error claim. The cited Cirac–Pérez-García–Schuch–Verstraete 2021 argument is not refuted. The source compresses two mathematical steps which the formal proof must keep separate: the B -region inverting and growing-back open-chain range reduction, and the boundary-closing comparison of the boundary matrices arising from the two boundary-crossing supports used in closing the boundary.

The open-chain part has been proved under the intended block-injective hypotheses, though not by reusing the older single-site intersection theorem. The last mathematical point is the comparison of the two boundary matrices at the periodic boundary. If that comparison is proved, the difference is only expository. If it reveals an additional mathematical obstruction, the conclusion should be revised accordingly.

References

- [CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.

⁹For traceability, this is the mathematical content recorded in issue #2368.